Steven Boada, Ph.D

Basic Information

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Professional Experience

Activision Publishing, Inc. — Boulder, Colorado

Senior Data Scientist, Gameplay — **January, 2024 - Present**

As part of the game science team, I work to make Call of Duty better through data. Currently, I am:

- Working with senior design leadership to understand how players interact with Call of Duty gameplay systems. This includes maps, modes, movement, weapons, and in-game communication among others.
- Leading initiatives to understand how game design affect topline KPIs, and suggest changes when appropriate.
- Supporting infrastructure improvements to better track or diagnose gameplay related issues.

Senior Machine Learning Engineer — **January, 2021 - 2024**

- Designed, built, maintained, and improved ML infrastructure. This includes tool developement like autoML, CI/CD (Jenkins, github actions), model tracking and management (MLflow) and orchestration (Airflow).
- Crafted and implimented policy around external data ingestion and data product retention for both my internal team and for sharing across the organization.
- Designed, built, and maintained near-realtime applications (via Spark streaming) to combat in-game cheating and other malicious behavior. Typical time-to-action in the low 10s of seconds.
- Lead the transition of ML infrastructure from AWS to GCP/GCS.
- Created ML models to provide insights into customer conversion, churn, and behavioral segmentation. This leveraged survival analysis, clustering, as well as tree-based and linear methods.
- Supervised junior team members to design and developed recommendation systems productionalized in both Modern Warfare 2 (2022) and Modern Warfare 3 (2023).

Insight Data Science — New York, New York

Fellow — **January, 2020 - 2021**

- Helped optimize the way NYC health inspectors perform restaurant inspections in order to reduce the time critical health violations remain unaddressed.
- Trained a random forest in Python to prioritize NYC restaurant inspections based on environmental variables and their past inspection histories and provided the results to NYC through an API deployed on AWS.
- Resulted in NYC inspectors identifying $\sim$2.5% more violations in the first half of an inspection window, leading to critical violations being discovered up to 7 days earlier than by the current approach implemented by NYC.

Dept. of Physics and Astronomy, Rutgers University — New Brunswick, New Jersey

Postdoctoral Research Associate — **September, 2016 - 2021**

- Designed and built parallelized pipelines to process and analyze TBs of astronomical imaging; producing calibrated, standardized data catalogs and rigorous results leading to 2 peer reviewed publications and several hundred hours of telescope time.
- Contributed to open source, astronomy-focused, Python projects through bug fixes and feature additions: see photometrypipeline, astLib, and easyGalaxy on GitHub as examples.

Skills

Machine Learning: Linear Models, Decision Trees, SVM, Clustering, Deep Learning, Survival Analysis
CI/CD: Jenkins, Airflow, Docker
Software and Computing: Open Source Contributor, Python, DataBricks, MLFlow, SQL, AWS/GCP, Spark, and other cloud computing applications
Leadership: Experience organizing and leading workshops and collaboration meetings, Teaching and mentoring junior team members, Eagle Scout.

Education

Texas A&M University, College Station, Texas

- Ph.D, Physics (astronomy focus), 2016

The University of Tennessee, Knoxville, Tennessee

- M.S., Physics (astronomy focus), 2009
- B.S., Physics, 2007